ECE 340: Semiconductor Devices

Lecture 23: P-N Junction Under Bias

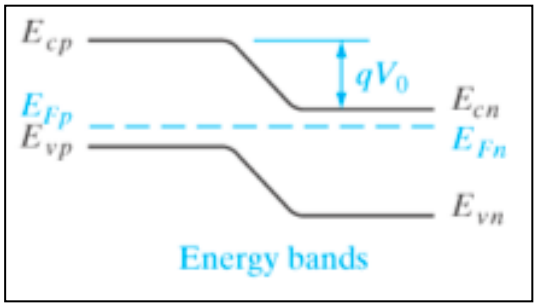
Prof. Hyunseok Kim

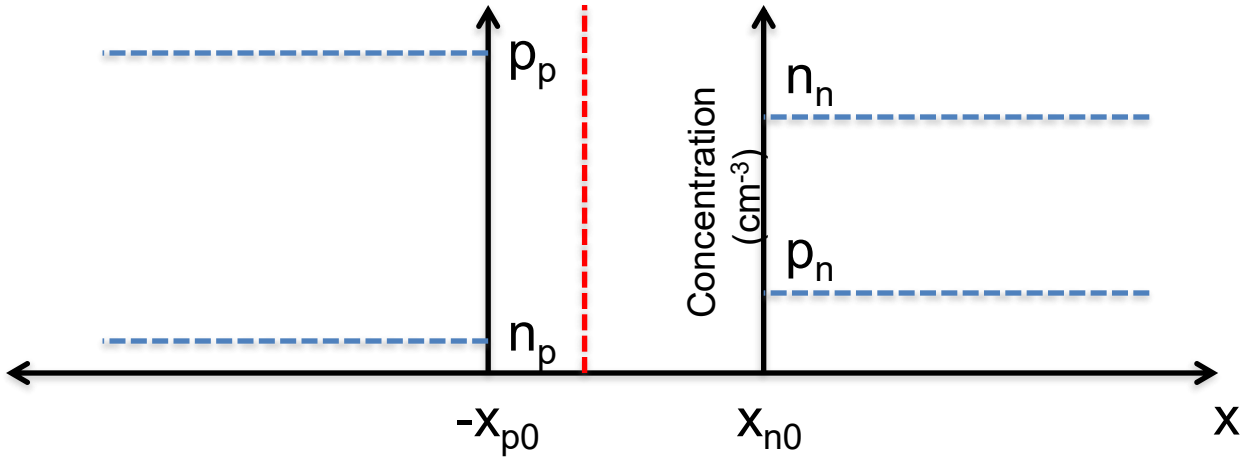
Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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(Recap) Carrier Concentrations vs. Built-in potential





$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$

$$p_p = p_n e^{qV_o/kT} \quad \text{or} \quad p_n = p_p e^{-qV_o/kT}$$

$$n_p = n_n e^{-qV_o/kT} \quad \text{or} \quad n_n = n_p e^{qV_o/kT}$$

(Recap) E Field and Contact Potential

Maximum Value of Electric Field:

$$\mathcal{E}_o = -\frac{q}{\varepsilon} N_d x_{n0} = -\frac{q}{\varepsilon} N_a x_{p0}$$

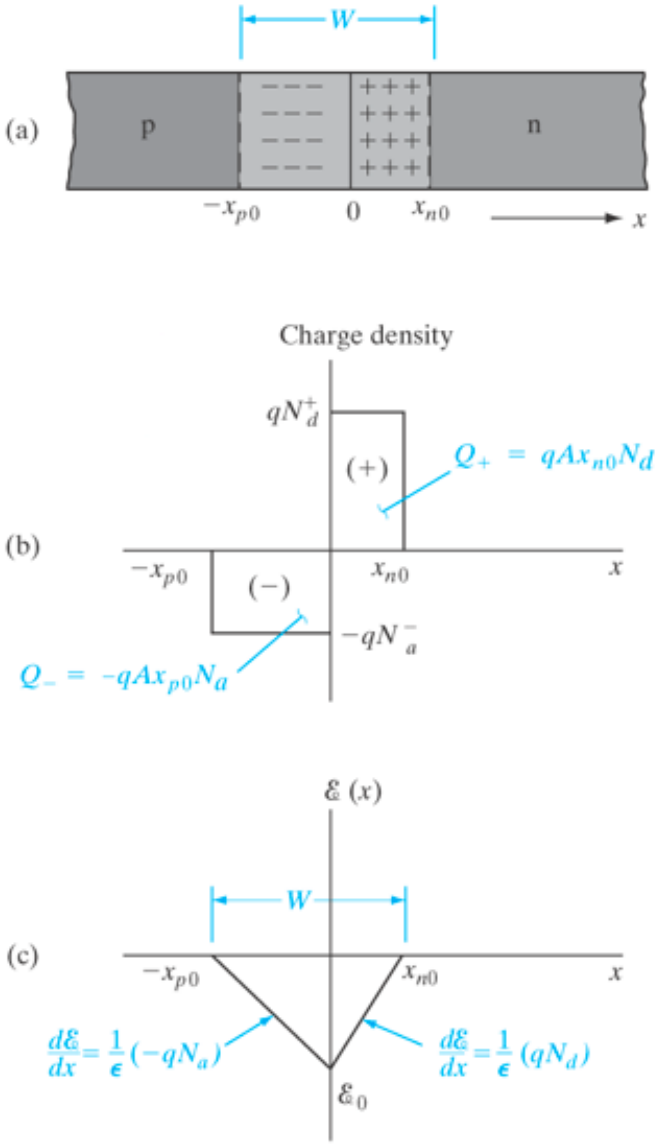
Relationship of Electric Field to Contact Potential:

$$\mathcal{E}(x) = -\frac{d\mathcal{V}(x)}{dx} \text{ or } -V_o = \int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

Since the electric field is the negative of the potential gradient at any point.

Recall that:

$$V_o = \frac{kT}{q} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \frac{n_n}{n_p}$$



(Recap) Calculation of Depletion Width

From the previous slide:

$$V_o = -\int_{-x_{po}}^{x_{n0}} \mathcal{E}(x)\,dx$$

So, the negative of the contact potential is the area under the $\mathcal{E}$(x) versus x triangle:

$$V_o = -\frac{1}{2}\mathcal{E}_o \cdot W = \frac{1}{2}\frac{q}{\varepsilon}N_d x_{no} W = \frac{1}{2}\frac{q}{\varepsilon}N_a x_{po} W$$

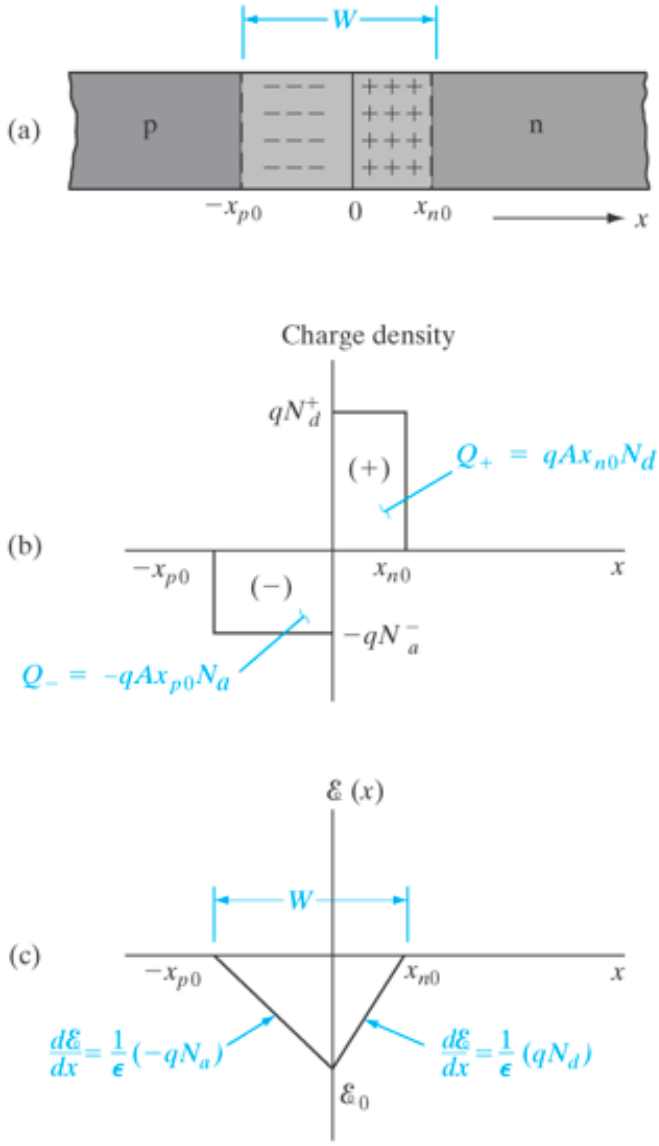
Balance of Charge Requirement: $Q_+ = |Q_-| \Leftrightarrow N_d x_{no} = N_a x_{po}$

$$x_{po} = x_{no}\frac{N_d}{N_a} \text{ and } W = x_{no} + x_{po} = x_{no}\left(1 + \frac{N_d}{N_a}\right),$$

$$\text{so } x_{no} = W\left(\frac{N_a}{N_a + N_d}\right)$$

$$\text{and } V_o = \frac{1}{2}\mathcal{E}_o W = \frac{1}{2}\frac{q}{\varepsilon}\frac{N_a N_d}{N_a + N_d}W^2$$

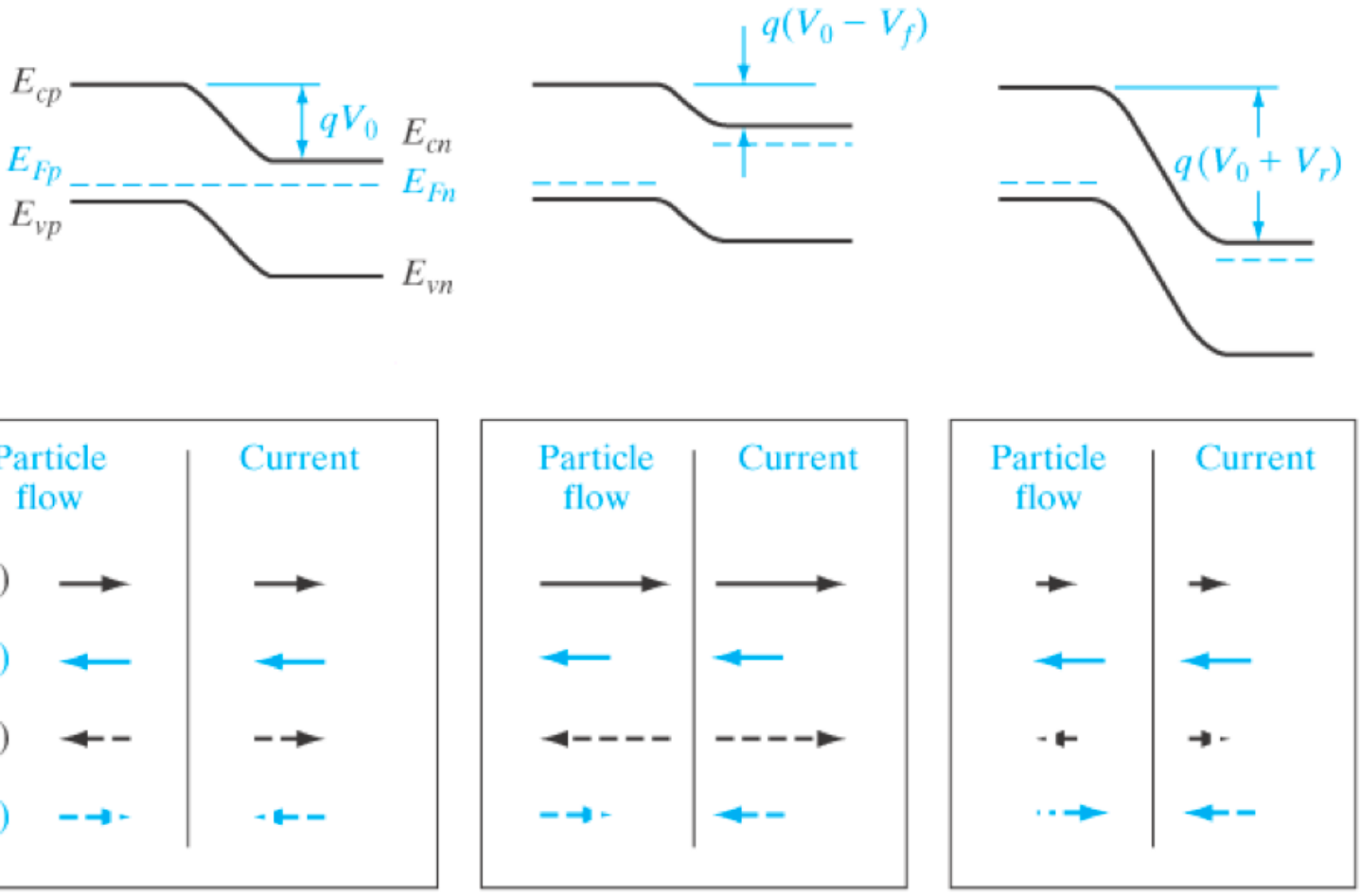
$$W = \left[\frac{2\varepsilon V_o}{q}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2} = \left[\frac{2\varepsilon V_o}{q}\left(\frac{1}{N_a} + \frac{1}{N_d}\right)\right]^{1/2}$$



pn Junctions Under Bias

A Few Comments

Electron & Hole Motion

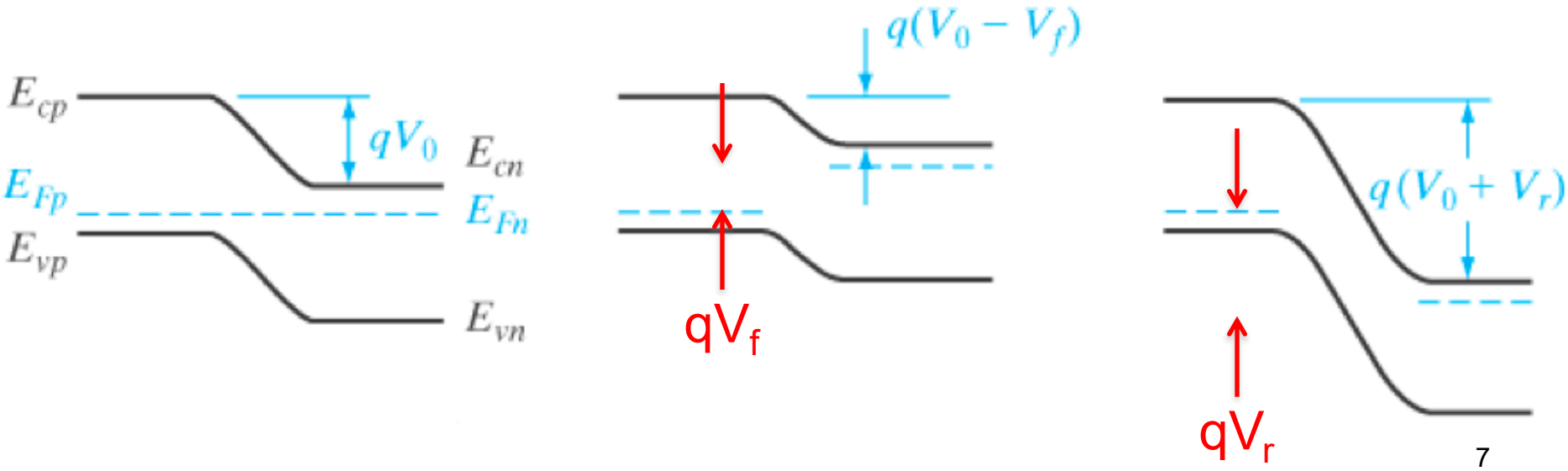


(1) Hole diffusion
(2) Hole drift

(3) Electron diffusion
(4) Electron drift

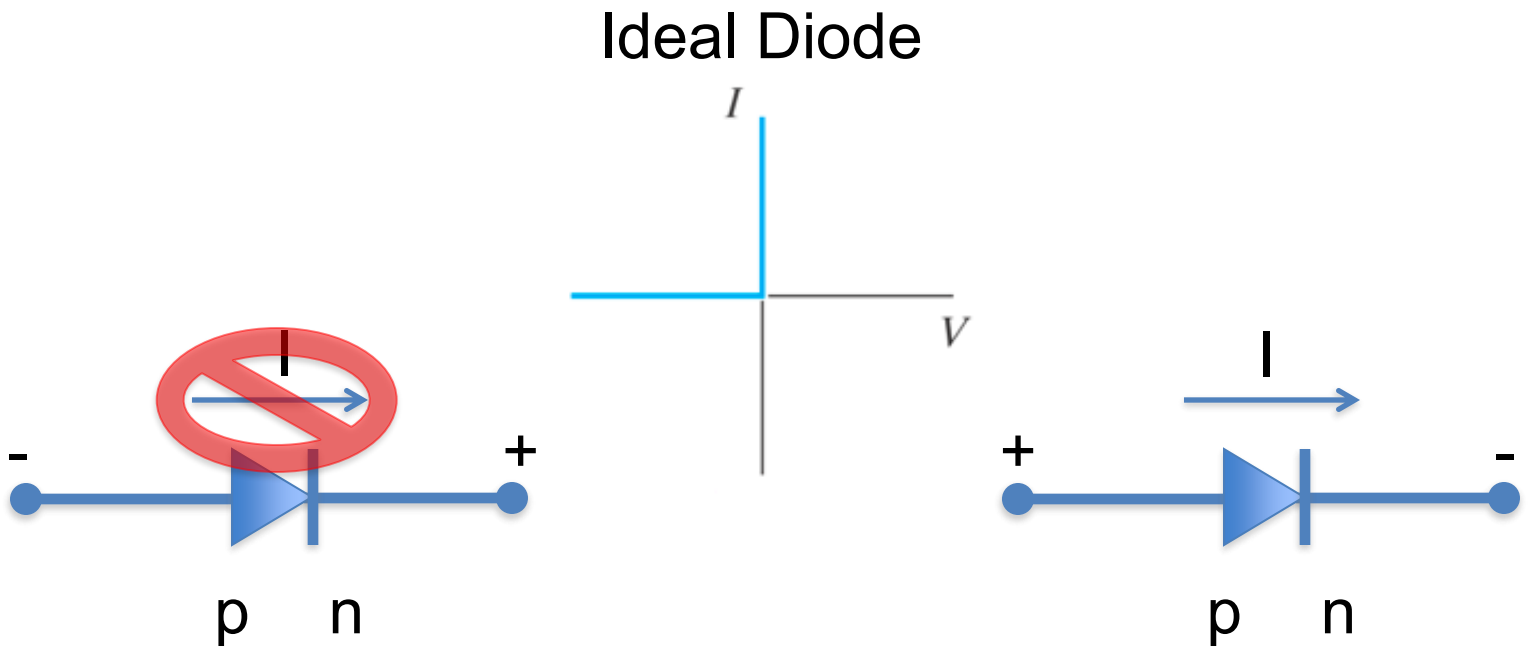
Bias and Fermi Level Separation

- When we apply voltage we are no longer in equilibrium
- The application of a voltage creates a difference in the electrochemical potential across the junction boundary
- The Fermi levels become separated by an energy of "q" times the applied voltage
- The band separation is $q(V_0-V)$



Terminology

- ***Rectifier:*** A device that passes current in one bias direction (positive to negative) and blocks the flow of current in the opposite bias direction (negative to positive)



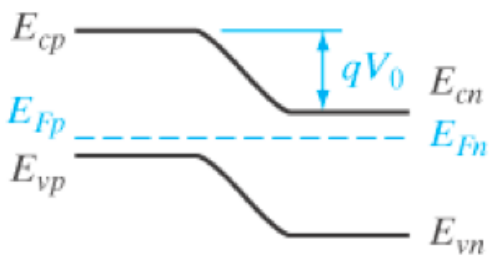
Carrier Injection

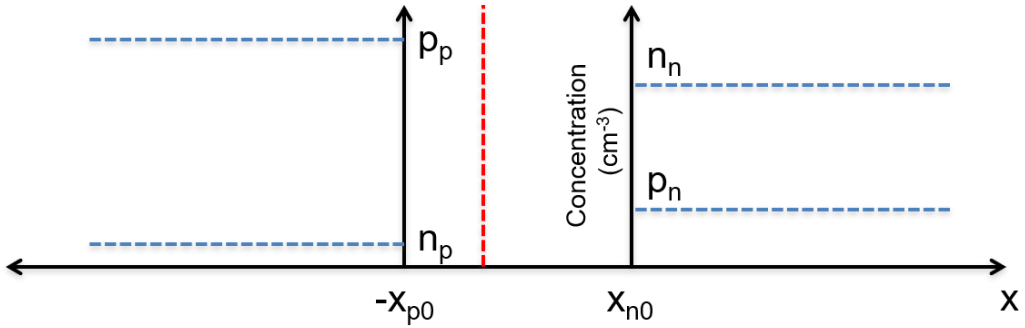
Minority Carrier Concentration

Recall, under equilibrium

$$n(x) = N_C \exp\left[-\frac{E_C(x) - E_{Fn}}{kT}\right]$$

$$\frac{n_p(-x_p)}{n_n(x_n)} = \exp\left[-\frac{E_C(-x_p) - E_C(x_n)}{kT}\right]$$





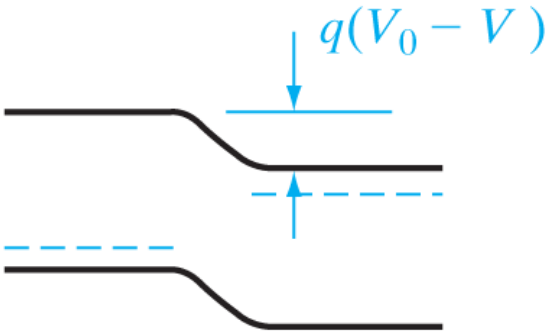
$$\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$$

$$p_p = p_n e^{qV_o/kT} \quad \text{or} \quad p_n = p_p e^{-qV_o/kT}$$

$$n_p = n_n e^{-qV_o/kT} \quad \text{or} \quad n_n = n_p e^{qV_o/kT}$$

Minority Carrier Concentration

Under bias (forward or reverse)



$$n(x) = N_C \exp\left[-\frac{E_C(x) - E_{Fn}}{kT}\right]$$

$$\frac{n_p(-x_p)}{n_n(x_n)} = \exp\left[-\frac{E_C(-x_p) - E_C(x_n)}{kT}\right]$$

Therefore,

$$n_p(-x_p) = n_n\, e^{-q(V_0 - V)/kT}$$

$$p_n(x_n) = p_p\, e^{-q(V_0 - V)/kT}$$

Minority Carrier Injection: Forward Bias

Recall in equilibrium: $\frac{p_p}{p_n} = \frac{n_n}{n_p} = e^{qV_o/kT}$

With a bias voltage V

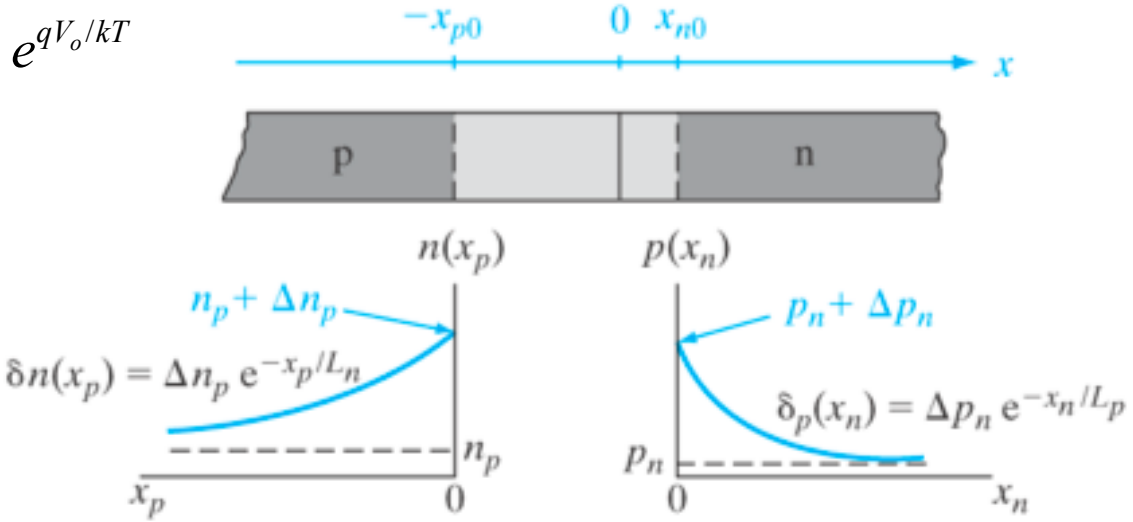
the barrier is reduced and:

$$\frac{p_p}{p_n} = \frac{p(-x_{po})}{p(x_{no})} = e^{q(V_o - V)/kT}$$

Under low-level injection,

neglect changes in **majority**

carrier concentration $\left(p_p = p(-x_{po})\right)$:



Equilibrium

$$\frac{\left(\frac{p_p}{p_n}\right)}{\left(\frac{p(-x_{po})}{p(x_{no})}\right)} = \frac{\left(\frac{p(-x_{po})}{p_n}\right)}{\left(\frac{p(-x_{po})}{p(x_{no})}\right)} = \left(\frac{p(-x_{po})}{p_n}\right)\left(\frac{p(x_{no})}{p(-x_{po})}\right) = \frac{p(x_{no})}{p_n} = \frac{e^{qV_o/kT}}{e^{q(V_o - V)/kT}} = e^{qV/kT}$$

Non-Equilibrium

Minority Carrier Injection: Forward Bias

From the previous slide:

$$\frac{p(x_{no})}{p_n} = e^{qV/kT} \quad \text{so:}$$

$$p(x_{no}) = p_n e^{qV/kT}$$

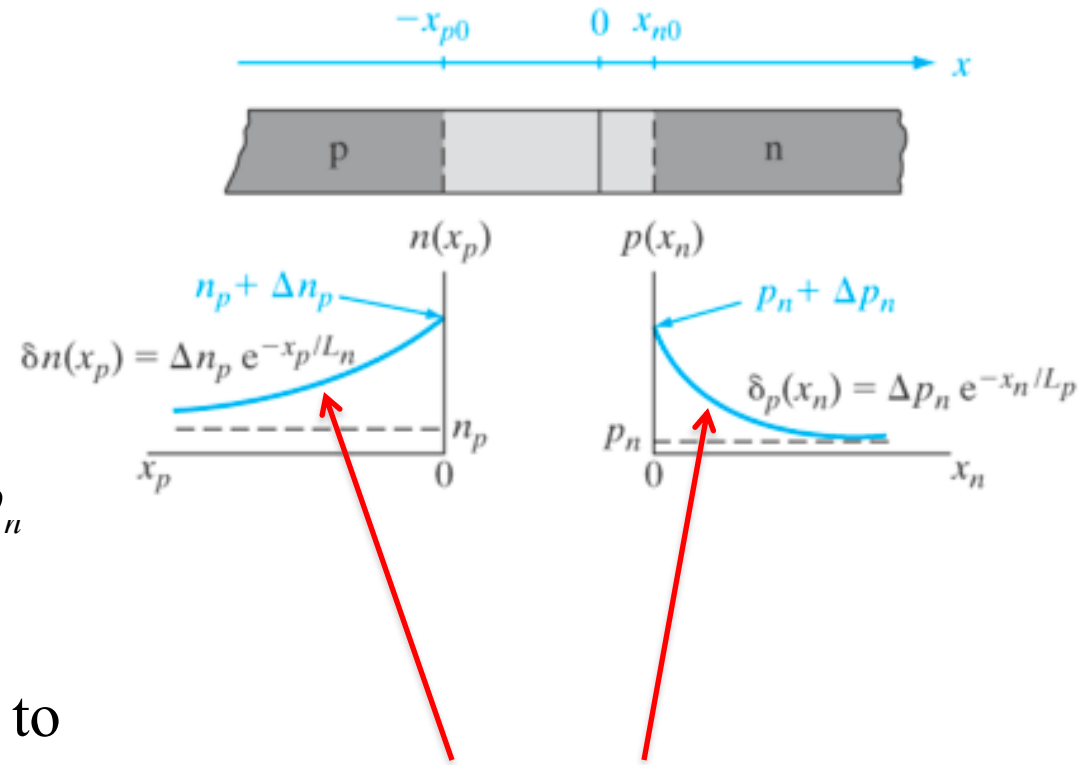
Therefore:

$$\Delta p_n = p(x_{no}) - p_n = p_n e^{qV/kT} - p_n$$

$$= p_n\left(e^{qV/kT} - 1\right)$$

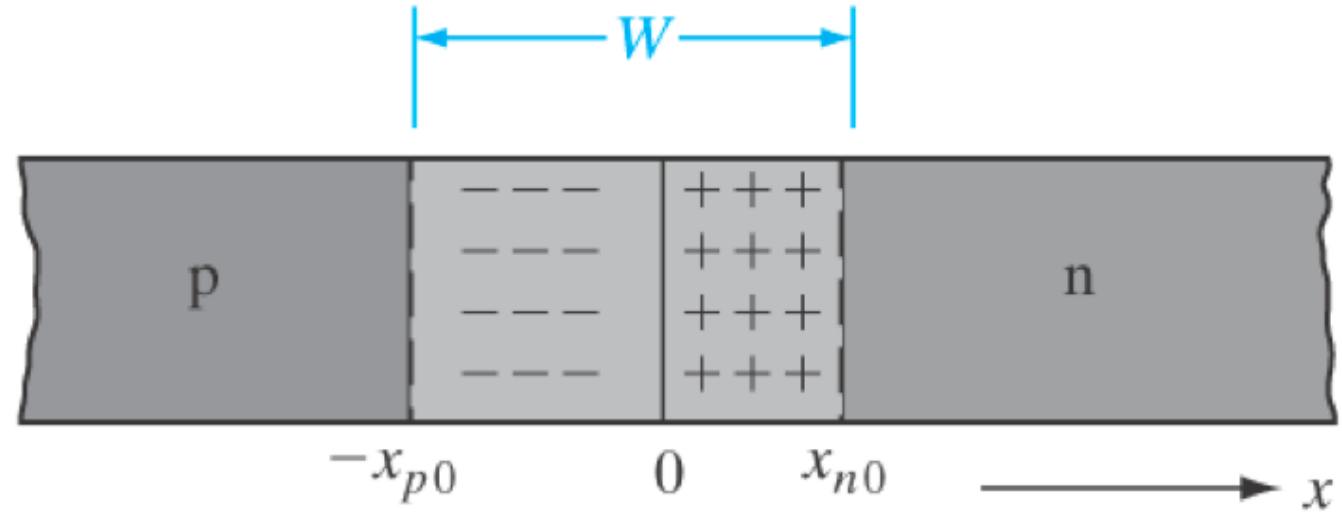
A similar approach can be used to derive the expression for Δn_p :

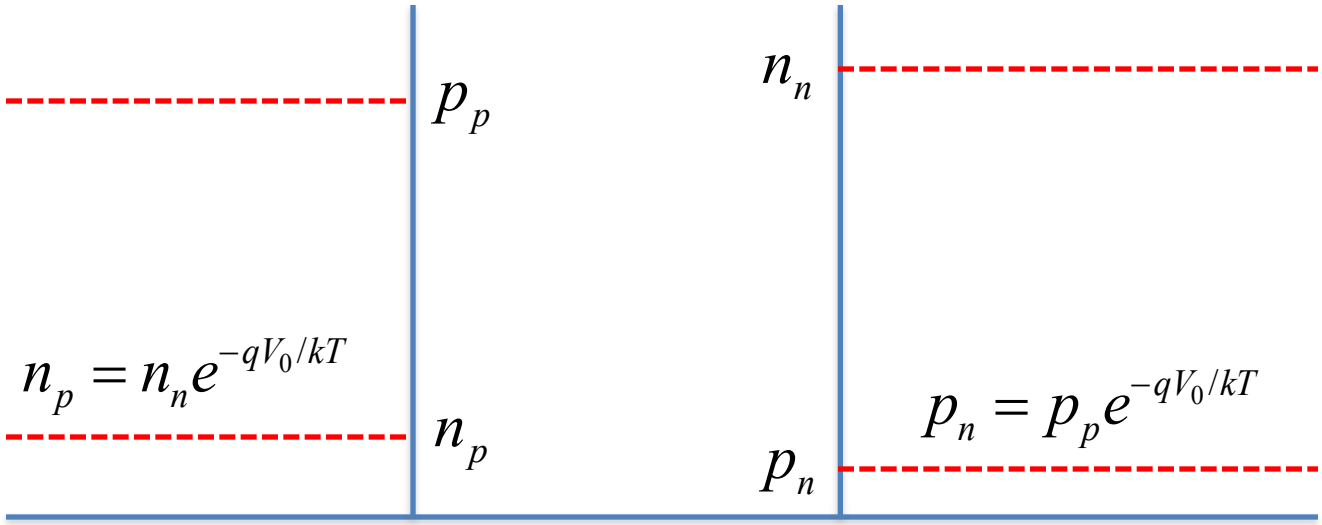
$$\Delta n_p = n(-x_{po}) - n_p = n_p\left(e^{qV/kT} - 1\right)$$



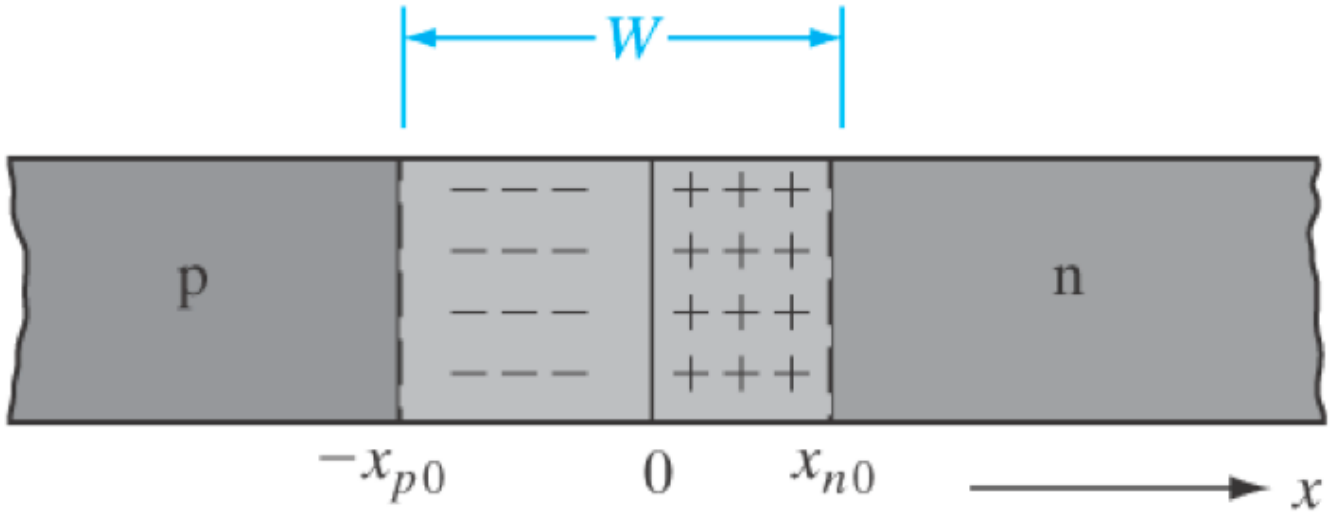
Exponential Decay due to Recombination

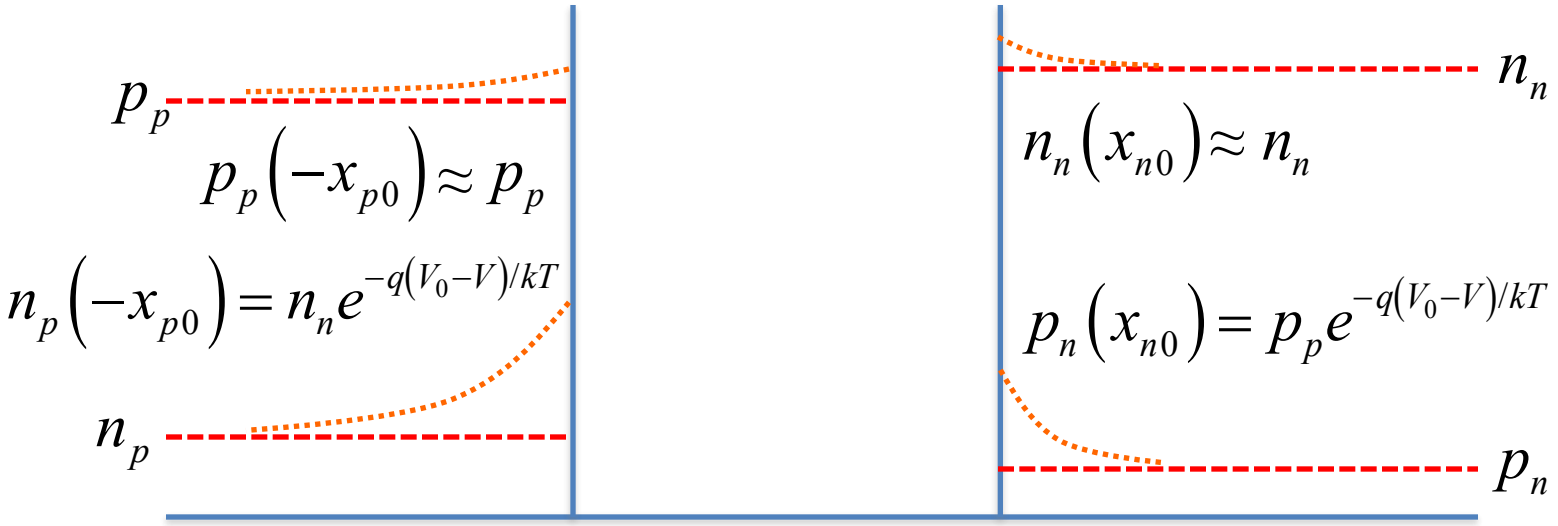
Equilibrium Carrier Concentrations



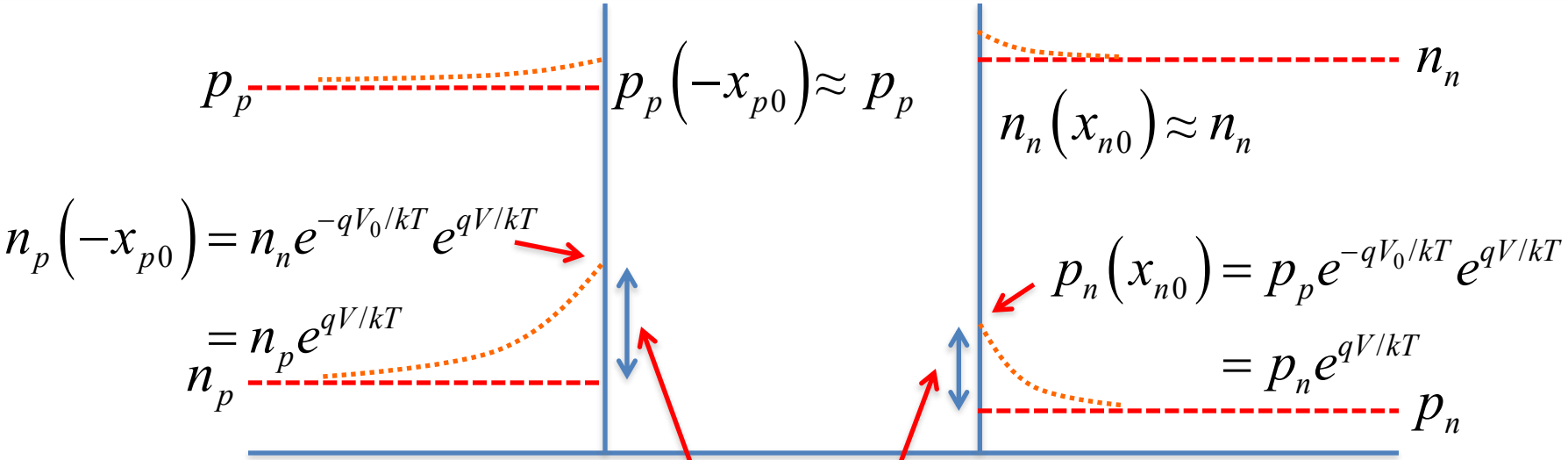


Forward Bias





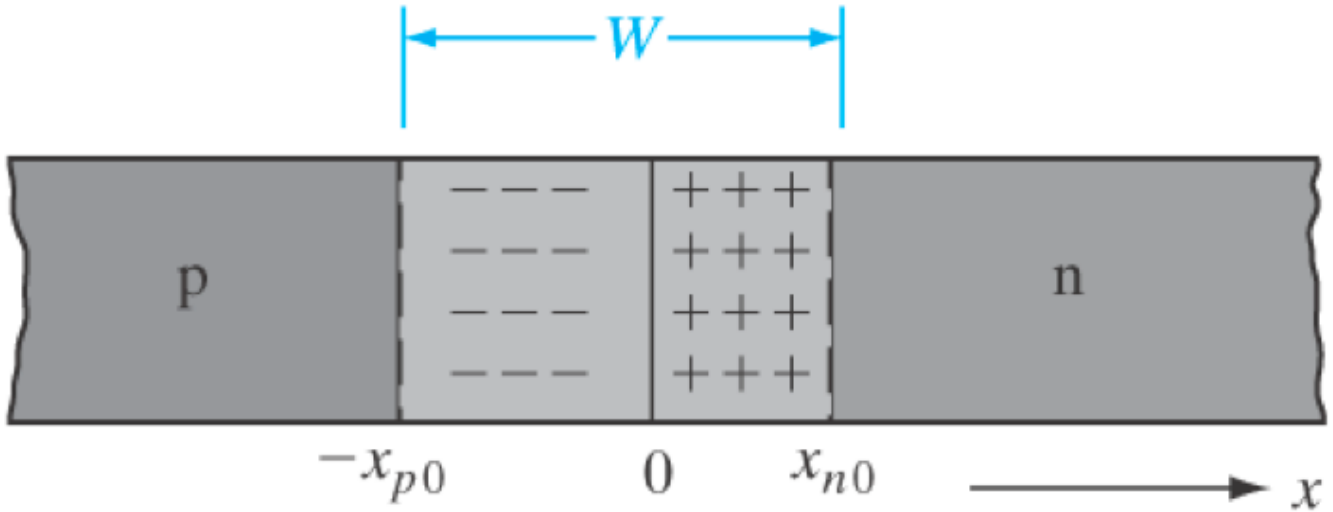
Forward Bias

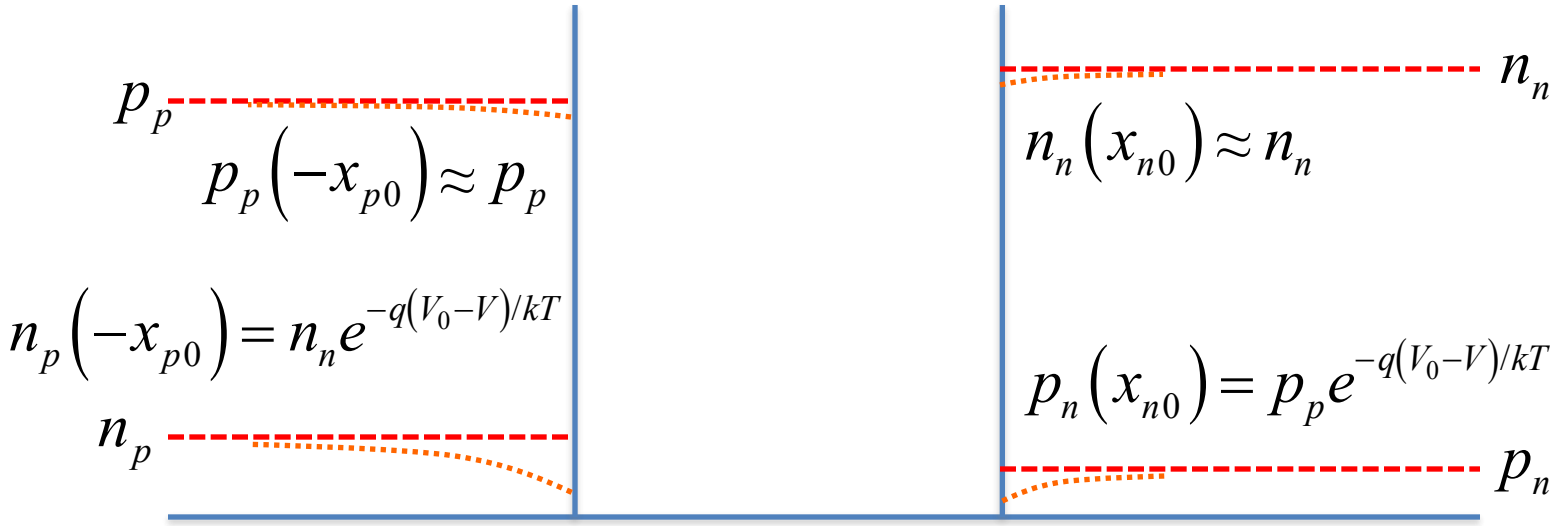


Space Charge Neutrality, Low - Level Injection :

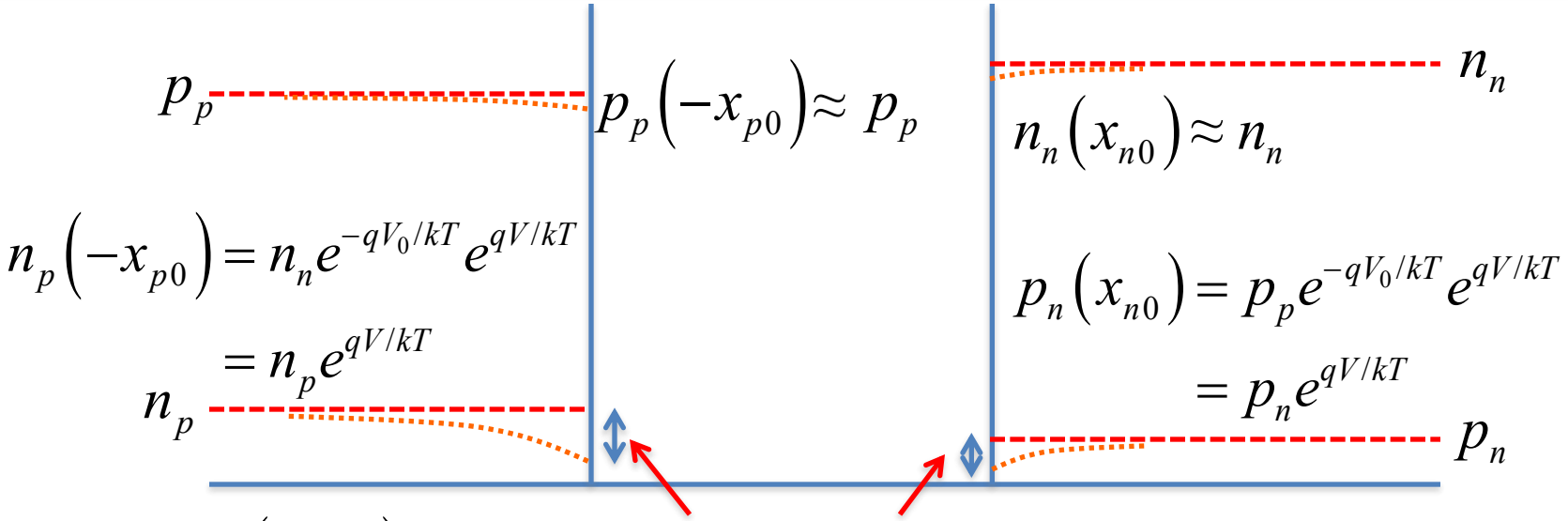
$$p_p(-x_{p0}) = p_p(-\infty) + \Delta n = p_p + n_p(e^{qV/kT} - 1) \approx p_p$$

Reverse Bias





Reverse Bias



$$n_p(-x_{p0}) - n_p(-\infty) = \Delta n$$

$$n_p e^{qV/kT} - n_p =$$

$$n_p\left(e^{qV/kT} - 1\right) =$$

$$-n_p \approx$$

$$\Delta p = p_n(x_{n0}) - p_n(\infty)$$

$$= p_n e^{qV/kT} - p_n$$

$$= p_n\left(e^{qV/kT} - 1\right)$$

$$\approx -p_n$$

Steady-State Carrier Injection and Diffusion Length

(Revisiting Earlier Material)

Diffusion Length: Steady State

Steady-State Diffusion Equations

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \frac{\delta p}{\tau_p}$$

In steady state, $\frac{\partial \delta p}{\partial t} = 0$

$$\therefore\ D_p \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{\tau_p} \text{ or } \frac{\partial^2 \delta p}{\partial x^2} = \frac{\delta p}{D_p \tau_p} \equiv \frac{\delta p}{L_p^2}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} - \frac{\delta n}{\tau_n}$$

In steady state, $\frac{\partial \delta n}{\partial t} = 0$

$$\therefore\ D_n \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{\tau_n} \text{ or } \frac{\partial^2 \delta n}{\partial x^2} = \frac{\delta n}{D_n \tau_n} \equiv \frac{\delta n}{L_n^2}$$

Hole Injection at x=0



(The injection mechanism is not specified.)

L_p is the average distance a hole travels before recombining.
L_n is the average distance an electron travels before recombining.

$$L_n \equiv \sqrt{D_n \tau_n} \text{ and } L_p \equiv \sqrt{D_p \tau_p}$$

Excess Hole Distribution and Current

Solution to Hole Diffusion Equation

$$\frac{d^2\delta p}{dx^2} = \frac{\delta p}{L_p^2} \qquad \frac{\partial \delta p}{\partial t} = 0$$

Solution:

$$\delta p(x) = C_1 e^{x/L_p} + C_2 e^{-x/L_p}$$

Boundary Condition:

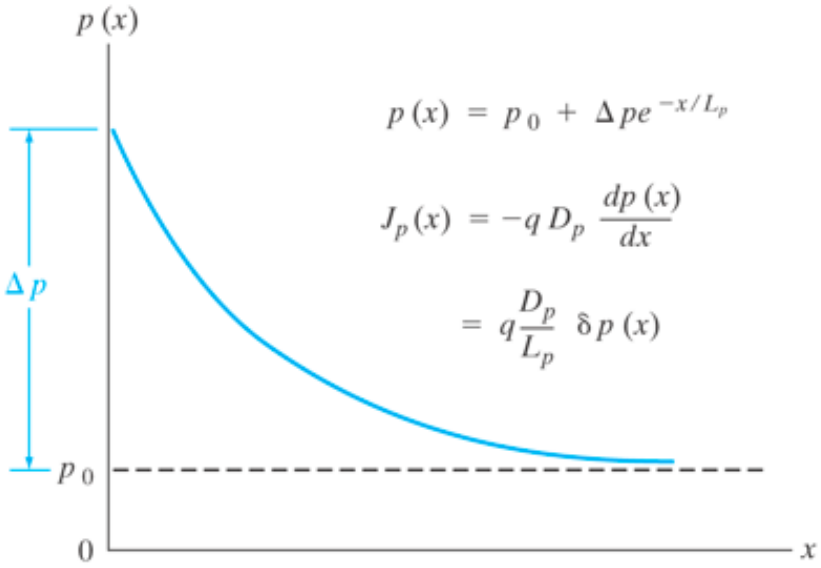
$$\delta p(x=0) = \Delta p = C_1 + C_2$$

$$\delta p(x=\infty) = 0 = C_1 e^{x/L_p}$$

$$C_1 = 0, \quad C_2 = \Delta p$$

$$\delta p(x) = \Delta p \, e^{-x/L_p}$$

Hole Injection at x=0



$$J_p(x) = -qD_p \frac{dp}{dx} = -qD_p \frac{d(p_o + \delta p)}{dx} = -qD_p \frac{d\delta p}{dx}$$

$$= q\frac{D_p}{L_p}\Delta p e^{-x/L_p} = q\frac{D_p}{L_p}\delta p(x)$$

Excess Electron Distribution and Current

Solution to Electron Diffusion Equation

$$\frac{d^2\delta n}{dx^2} = \frac{\delta n}{L_n^2} \qquad \frac{\partial \delta n}{\partial t} = 0$$

Solution:

$$\delta n(x) = C_1 e^{x/L_n} + C_2 e^{-x/L_n}$$

Boundary Condition:

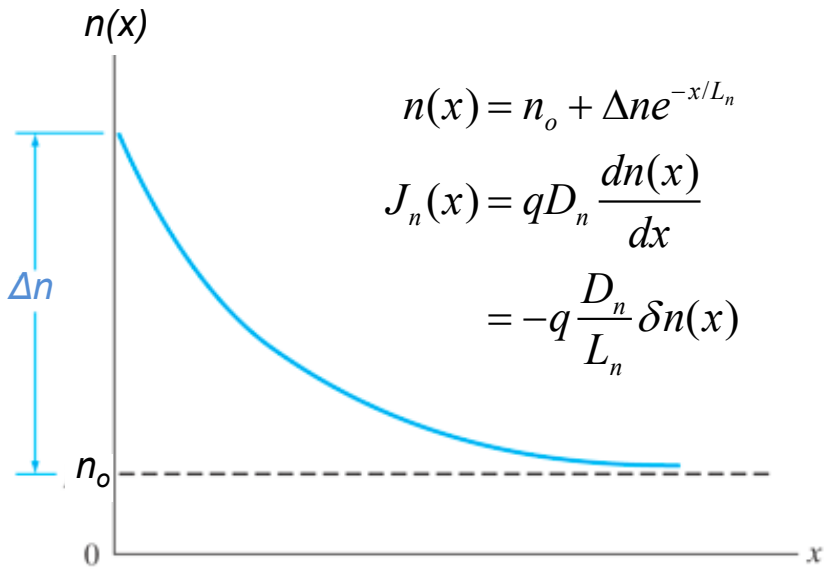
$$\delta n(x=0) = \Delta n = C_1 + C_2$$

$$\delta n(x=\infty) = 0 = C_1 e^{x/L_n}$$

$$C_1 = 0, \quad C_2 = \Delta n$$

$$\delta n(x) = \Delta n \, e^{-x/L_n}$$

Electron Injection at x=0



$$n(x) = n_o + \Delta n e^{-x/L_n}$$

$$J_n(x) = qD_n \frac{dn(x)}{dx} = -q\frac{D_n}{L_n}\delta n(x)$$

$$J_n(x) = qD_n \frac{dn}{dx} = qD_n \frac{d(n_o + \delta n)}{dx} = qD_n \frac{d\delta n}{dx}$$

$$= -q\frac{D_n}{L_n}\Delta n e^{-x/L_n} = -q\frac{D_n}{L_n}\delta n(x)$$

Electron Diffusion in –x Direction

Solution to Electron Diffusion Equation

$$\frac{d^2\delta n}{dx^2} = \frac{\delta n}{L_n^2} \qquad \frac{\partial \delta n}{\partial t} = 0$$

Solution:

$$\delta n(x) = C_1 e^{x/L_n} + C_2 e^{-x/L_n}$$

Boundary Condition:

$$\delta n(x=0) = \Delta n = C_1 + C_2$$

$$\delta n(x=-\infty) = 0 = C_2 e^{-x/L_n}$$

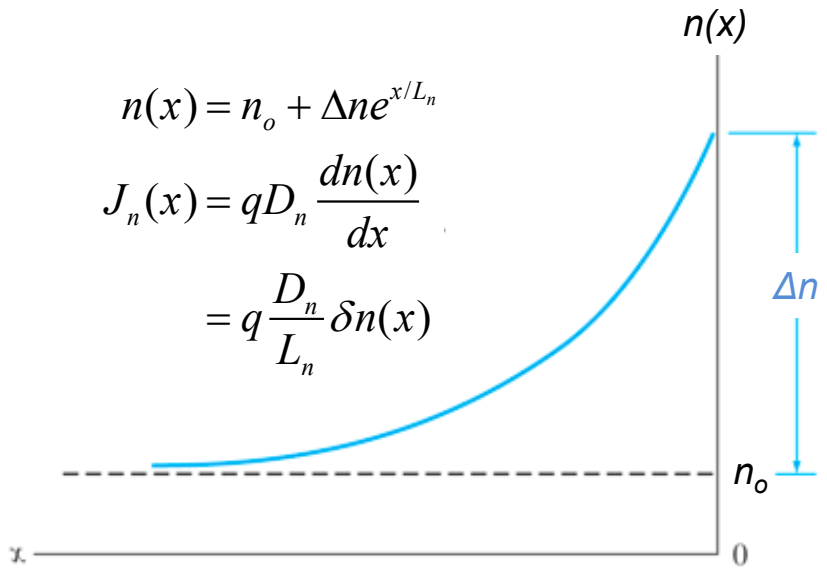
$$C_2 = 0, \quad C_1 = \Delta n$$

$$\delta n(x) = \Delta n \cdot e^{x/L_n}$$

Electron Injection at x=0

$$n(x) = n_o + \Delta n e^{x/L_n}$$

$$J_n(x) = qD_n \frac{dn(x)}{dx} = q\frac{D_n}{L_n}\delta n(x)$$



$$J_n(x) = qD_n \frac{dn}{dx} = qD_n \frac{d(n_o + \delta n)}{dx} = qD_n \frac{d\delta n}{dx}$$

$$= q\frac{D_n}{L_n}\Delta n e^{x/L_n} = q\frac{D_n}{L_n}\delta n(x)$$